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Economic freedom and the quality of education

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Abstract

This paper finds robust evidence that economic freedom improves the quality of education. This is probably mainly because economic freedom incentivizes parents to invest in high-quality education for their children and helps them to do so. It also incentivizes and helps both governments and private providers to deliver high-quality education. The paper uses two different indicators of educational quality: PISA scores and the World Bank's harmonized test scores. The magnitudes of the estimated effects of economic freedom on both indicators are substantial. They are even larger once the indirect impact of economic freedom via GDP per capita and, to a lesser extent, once the indirect impact via government education expenditure is taken into account. The paper uses data on up to 49 countries for PISA scores and up to 137 countries for World Bank scores. It accounts for the endogeneity of economic freedom and controls for all major determinants of educational quality.

1 | INTRODUCTION

This paper studies whether the extent of countries' economic freedom affects the quality of the education they achieve. As section 2 explains in detail, there are several reasons why there is likely to be a positive effect. Most importantly, economic freedom incentivizes parents to invest in high-quality education for their children and helps them to do so. It also incentivizes and helps both governments and private providers to deliver high-quality education. Indeed, using data from a large number of countries and two different indicators of educational quality, we find robust evidence that economic freedom improves educational outcomes.

Our paper is important for two reasons. First, prior research has shown that the quality of education is crucial for both students and the economy. At the individual level, various studies find that the quality of the education

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children receive is critical for their later earnings (e.g., Hanushek et al., 2015; Hanushek et al., 2017). Martellini et al. (2024) report that college graduate quality predicts the share of a college's students who become inventors, engage in entrepreneurship, and become top executives. At the macro level, Hanushek and Woessmann (2011a, 2012a, 2012b, 2015, 2016) document in a string of publications that the cognitive skills of the population are a key driver of GDP growth. Schoellman (2012) finds that cross-country differences in education quality are important in accounting for international differences in output per worker. Thus our result that economic freedom improves educational achievement has important implications for both students and macroeconomic performance. Consequently, it also has important implications for public policy, which we discuss in the final section of our paper.

The second reason why our paper is important is that the effect of economic freedom on the quality of education has not been studied so far. This is despite the fact that the sources of international differences in student achievement have already been analyzed in a vast number of studies (for surveys, see Hanushek & Woessmann, 2011b, Woessmann, 2016). Most of them, however, focus either on the characteristics of school systems such as governmental education expenditure, class size, and teacher quality, or on the role of parents and families. There is also a large, separate literature on the effects of economic freedom (for surveys, see Hall & Lawson, 2014, Lawson, 2022). Although papers in this literature employ many different outcome variables such as GDP growth, unemployment, and inequality, very few of them tackle the effects on education (section 3). Our paper therefore addresses a significant gap in the literature.

The two indicators of educational quality we use are the OECD's PISA scores and the World Bank's harmonized test scores. A major advantage of PISA is that it provides not only achievement data but also a rich set of background information on students, their families, and schools that we can use as controls. The major advantage of the World Bank's harmonized test scores, which are based on several multi-country testing programs, is that they are available for many more countries. Specifically, whereas our PISA regressions cover 49 countries at most, our regressions using World Bank scores cover up to 137 (for lists of countries, see the online Supporting Information section). The downside of the World Bank scores is that they are country-level indicators and do not come with any information on students and schools. We use both these indicators to check whether our results are robust.

Our paper is organized as follows. The next section argues theoretically why economic freedom is likely to improve the quality of education. Section 3 briefly summarizes the previous literature studying the link between economic freedom and education. Section 4 explains how we measure and instrument economic freedom. While section 5 presents our analysis using PISA scores, section 6 presents our analysis using World Bank scores. Section 7 studies the ways in which economic freedom is likely to affect educational performance. Section 8 summarizes our results and discusses its policy implications.

2 | THEORETICAL CONSIDERATIONS

Economic freedom is likely to improve the quality of education for several reasons. First, it enhances the gains from educational investments. A modest level of taxation, a key element of economic freedom, implies that individuals are able to achieve comparatively high net returns to education. Monetary stability, another element of economic freedom, ensures that individuals' returns to education are not diluted by inflation. A pro-competitive regulation of labor markets enables individuals to enter professions where their educational returns are the highest. A light regulatory burden on product markets and low barriers to international trade, two further elements of economic freedom, foster specialization and economic exchange, enhancing the gains from educational investments as well. Secure property rights and the rule of law not only prevent the state from expropriating people's earnings, they additionally ensure that the state and the economy function in a predictable manner, helping individuals to make the best use of their human capital.

The various elements of economic freedom operate jointly. For example, a strong rule of law and secure property rights are a precondition for both governmental regulations and international trade to work smoothly. Another example is monetary stability; it secures people's real net returns to educational investments jointly with light taxation.

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As economic freedom enhances the returns to educational investments, it incentivizes parents to make their children more aware of the importance of learning and to induce them to put more effort into it. It also incentivizes them to invest in high-quality education for their children, e.g., by funding private tuition and choosing the best schools. Previous research shows that, in both developed and developing countries, most parents attach great importance to the quality of education their children receive and that many are prepared to pay for good quality (Feldmann, 2020). Economic freedom also makes parents more likely to request head teachers to raise educational standards and motivates them to elect political parties that promise to improve the quality of schools. In short, it incentivizes them to strive for better educational quality because a better quality raises their children's future earnings. To improve their electoral prospects, politicians in turn have an incentive to meet parents' demands, primarily by raising standards in government-run schools.

Second, as economic freedom facilitates the operation of credit markets, it makes it easier for parents to take out loans to fund the best possible education, thereby probably raising their children's academic achievements. Key elements of economic freedom relevant here are secure property rights (including the right to use property as collateral) and a pro-competitive regulation of credit markets. Education entails a large upfront fixed cost while the return in the form of higher wages accrues later throughout working life. Parents therefore often need to take out loans to invest in a quality education for their children (e.g., Mimoun, 2008). The easier it is to use their property as collateral and the more competitive credit markets are, the easier and cheaper it is for them to do so (e.g., Sun & Yannelis, 2016).

Third, as a fundamental characteristic of economic freedom is freedom of choice, a country with more economic freedom is likely to provide parents with more choice between schools, including through a larger scope for privately operated schools. As the quality of instruction is key for many parents when choosing a school for their child and as parental choice increases competition between schools, more school choice can incentivize governments as well as private providers to improve educational quality (e.g., Belfield & Levin, 2002; West & Woessmann, 2010).

Finally, as economic freedom has been shown to boost economic growth and income per capita (e.g., Bennett et al., 2017; Feldmann, 2005; Grier & Grier, 2021), it increases resources for educational investments, both for parents (via higher incomes) and for governments (via higher tax revenue). Parents can use these resources to fund private tuition, send their children to private schools, or move to catchment areas of better public schools. Governments can use the additional tax revenue to increase their spending on education – for example, by investing in school buildings or instructional material, or by hiring more teachers and improving their training.

3 | RELATED LITERATURE

Only two previous papers resemble ours. The first is by King et al. (2012). While we study the determinants of student test scores, they study the determinants of private returns to education. Using data on 86 developing countries, they find the returns to be substantially higher in economically free countries. According to their regression results, the benefits go to men as well as women. The second similar paper is by Feldmann (2021), who reports that people in economically freer countries care more about education. Moreover, he documents a positive correlation between economic freedom and the World Bank's harmonized test scores but does not attempt to estimate the effect of the former on the latter.

All other previous papers studying the impact of economic freedom on education use measures of educational "quantity" rather than quality, mostly schooling rates. Schooling rates and educational quality do not necessarily evolve in lockstep. Le Nestour et al. (2022) document large differences between the two in the developing world: whereas schooling substantially increased, educational quality stagnated or declined in the second half of the 20th century.

The first paper studying the effect of economic freedom on educational "quantity", by Dawson (1998), only briefly touches upon the issue as it is mainly concerned with the determinants of economic growth and physical

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investment. However, it also reports the results of a few regressions to explain the secondary enrollment rate. In these regressions, which use data on 84 countries, Dawson (1998) finds a positive association with change in economic freedom.

Stroup (2007) reports a positive effect of economic freedom on the adult literacy rate in a sample of 68 countries and on the percentage of the population attaining at least a fifth-grade education in a sample of 24 countries. Using data from 109 countries, Feldmann (2017) finds that economic freedom has a substantial positive effect on both the overall and the female secondary enrollment rates. Aixalá and Fabro (2009) report bidirectional Granger causality between economic freedom and primary school enrollment in a sample of 112 countries.

Several other papers focus on the effect of education on economic freedom, rather than vice versa. For example, using data on up to 109 countries Murphy and O'Reilly (2019) find a positive effect. Their measure of education is the Penn World Table's index of human capital per person, which is based on years of schooling and returns to education. Another example is Jones and Potrafke (2014), who use data on up to 130 countries and find that cognitive skill measures have a positive effect on property rights protection, a key element of economic freedom.

4 | MEASURING AND INSTRUMENTING ECONOMIC FREEDOM

As our variable of interest, we use the 'Economic Freedom of the World (EFW)' summary index, scaled to range from 0 (least free) to 1 (most free). (For definitions, sources, and descriptive statistics of all variables, see the online Supporting Information section.) The EFW index is the most widely used index of economic freedom and is generally considered to be the best because it provides a comprehensive measure of the degree to which countries' institutions and policies are consistent with economic freedom. Moreover, it uses data from reputable external sources (such as the World Bank and the IMF), relies on objective data as far as possible, and covers a large number of countries. The EFW index comprises 42 distinct variables that are organized into five areas: the size of government, legal system and property rights, sound money, freedom to trade internationally, and regulation of credit, labor, and business (Gwartney et al., 2022). The summary ratings of the index are the arithmetic means of the area ratings.

For two reasons, we use the summary index rather than the area ratings. First, the work of eminent scholars over the past few centuries has led to the insight that all five areas are important and interrelated aspects of economic freedom. Major examples include Locke (1689 [1988]), Smith (1776 [1981]), Mill (1859 [1989]), Hayek (1960) and Friedman (1962). Indeed, the creators of the EFW index were careful in taking all key dimensions of economic freedom identified by these scholars into account (Block, 1990; Easton & Walker, 1992). The second reason why we use the summary index is because, as explained in section 2, all areas of economic freedom are likely to affect educational achievement jointly. Still, to examine the relative importance of each area, we substitute the area ratings for the summary index in section 7.

We instrument 'economic freedom' to extract its exogenous component. There are three reasons for doing so. First, as some previous papers suggest, better-educated people may prefer more economic freedom (reverse causality bias). Second, our models do not account for all determinants of student achievement (omitted variable bias). This is particularly the case when using the World Bank's harmonized test scores. Third, our variable of interest may measure the true extent and characteristics of economic freedom with some error (measurement error).

In most of our models using PISA data, the instruments are lagged differences of the instrumented variable covering the years t–1 to t–6. Obviously, changes in economic freedom over the previous six years affect its current level. Moreover, it is plausible to assume that those changes affect current educational performance not directly but only through the current extent of economic freedom. More generally, the insight that lagged differences of an endogenous variable can be valid instruments stems from, inter alia, the development of the system GMM estimator (Arellano & Bover, 1995). In Monte Carlo simulations, Arellano and Bover (1995), and Blundell and Bond (1998) show that using instruments in first differences for equations in levels can lead to large improvements in efficiency and

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consistency, compared with the original difference GMM estimator. As the PISA data is not a panel dataset, GMM estimation is not an option here, so we use two-stage least squares (2SLS) instead.

In our regressions with World Bank test scores and in one alternative specification with PISA data, we use as instruments lagged levels of economic freedom from the previous two years. We do so because, in these cases lagged differences fail to pass some of the IV validity tests. The rationale for using lagged levels of economic freedom is that they are likely to affect current educational performance only through the current level of economic freedom. That is, while a country's degree of economic freedom in the recent past clearly influences its current degree of economic freedom, it probably affects its current educational performance only through its current degree of economic freedom. Various econometricians have argued that using lagged levels of an endogenous variable as instruments can serve to mitigate the endogeneity problem (Anderson & Hsiao, 1981; Reed, 2015; Todd & Wolpin, 2003). Indeed, this instrumentation strategy is not uncommon in applied economics (e.g., Durlauf et al., 2008; Layard & Nickell, 1986).

Our instruments, both lagged differences and lagged levels of economic freedom, have three more advantages. First, they stand in a close economic relationship to the instrumented variable. As Acemoglu (2005) notes, there is little justification for using instruments that are only loosely related to the endogenous variable. Second, our instruments have not been used to instrument other variables. As Bazzi and Clemens (2013) point out, an instrument can be valid for only one endogenous variable. Third, our instruments pass all tests for instrument validity (see below).

A further reason for using internal instruments is that valid external instruments for economic freedom are lacking. Only a few previous papers have used any. While Bennett and Nikolaev (2019) apply an index of the historical prevalence of infectious diseases, Bennett and Nikolaev (2017) employ "the suitability of land for growing wheat relative to sugarcane". The latter variable was constructed and used by Easterly (2007) as an instrument for income inequality. Faria and Montesinos (2009) adopt instruments that have been used in the growth literature, such as legal origins. Therefore, the instruments used by Bennett and Nikolaev (2017) and Faria and Montesinos (2009) are probably invalid because they have been used to instrument other variables (Bazzi & Clemens, 2013). In any case, we tried these instruments as well as the disease instrument and a multitude of other variables and found none of them to pass the IV validity tests (results not reported here).

5 | ANALYZING PISA SCORES

5.1 | Data and methodology

As our first measure of educational quality, we use PISA mathematics scores, divided by 100. PISA tests random samples of 15-year-old students in three-year intervals, independent of the educational track attended or the school year they are in (OECD, 2019). The OECD's high sampling and data-quality standards ensure that each country's results are representative. The PISA math test asks students to apply their mathematical knowledge to solve real-world problems. Math scores are generally considered to be most readily comparable across countries, in contrast to other subjects such as reading, which is to some extent language-specific (Hanushek et al., 2022).

In our PISA regressions, we control for all major determinants of educational performance that have been considered in the literature. Additionally, we introduce several new control variables that could plausibly have an effect as well. Our country-level controls include a dummy for Confucian heritage societies, GDP per capita, the share of government funding for schools, the share of private schools, the share of students subject to central exit exams, and an index of school autonomy. Our school-level controls are the share of certified teachers, a dummy for a shortage of educational material, two school location dummies, and a school's number of students. Our individual-level controls cover sex, age, migration status, parental education and occupation, books and computers at home and language spoken at home. In our robustness checks, we add country-level variables measuring political freedom, preferences for patience and risk-taking, air pollution, output gap, unemployment, and the stock of FDI in the economy.

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As explained in section 2, economic freedom may affect student achievement through some of these control variables – specifically, through GDP per capita, private schools, and variables measuring school and family resources. Section 7 explores these indirect effects in some detail. Here, we include these controls. For two reasons, it would be inappropriate to omit them. First, they are not exclusively determined by economic freedom. Rather, they are co-determined by a range of other factors as well. For example, GDP per capita is largely determined by the amount of resources and their productivity. The availability of school and family resources and the presence of private schools are mostly determined by factors such as GDP per capita, politics, culture, and, with respect to family resources, individual circumstances and preferences. Second, these controls can be expected to directly affect educational performance, both from a theoretical point of view and because of previous empirical research. For these reasons, excluding them would lead to omitted variable bias.

Next, let us briefly explain the new control variables we use. Our first is 'Confucian heritage society'. In these societies – i.e., countries such as China, Japan, South Korea, and Vietnam – education has been held in high regard for centuries (Yang, 1993). Nowadays people still attach great importance to it (Starr, 2012; To, 1993). Building on this attitude, East Asian governments made education a key element of their development strategies, investing heavily in this area (Page, 1994). Additionally, parents support their children with out-of-school tuition (Zhang & Yamato, 2018). In line with Confucian values and principles, educational practice is characterized by hard work and relentless assessment. Given this background, the stellar performance of these countries in PISA and other international achievement tests should have come as no surprise (Jerrim, 2015).

In our first robustness check, we add 'political freedom'. We do so for two reasons – first, to ensure that economic freedom does not proxy for it, and second, to check whether political freedom exerts an influence of its own on the quality of education. Several studies find politically free countries to have higher enrollment rates (e.g., Eterovic & Sweet, 2014; Lake & Baum, 2001); their educational investments might also lead to better educational outcomes.

Several recent papers study the influence of cultural factors on student achievement (e.g., Figlio et al., 2019; Mendez, 2015). A major study is by Hanushek et al. (2022), who focus on patience and risk-taking – two preference components that reflect the long-term and risky nature of educational investments. They find that patience is positively and risk-taking negatively associated with test scores. Following their study, we add both variables in our second robustness check.

The impact of air pollution on students' test scores has been studied in recent years in a few countries, consistently finding that more pollution reduces them (for a survey, see Gartland et al., 2022). Whereas these studies use school- or district-level data, we employ an indicator measuring air pollution at the country level in our third robustness check.

In our fourth and fifth robustness checks, we add 'output gap' and 'youth unemployment rate', respectively. Recessions and unemployment could adversely affect student performance by limiting parents' and governments' resources to support student learning. On the other hand, during recessions and periods of high youth unemployment, parents may put additional pressure on their children to learn, and students may be more intrinsically motivated, both of which could improve their test results. Using data on Spain, Petrongolo and San Segundo (2002) find that youth unemployment increases the demand for education.

In our final robustness check, we add 'FDI stock'. A large stock of FDI could be associated with better educational quality because foreign-owned firms frequently require comparatively more workers with higher skills (Saurav et al., 2020).[1]

In our baseline specification, we use data from the 2018 wave and run two-stage least squares regressions of the following form:

Second stage:

[1]'FDI stock' is conceptually different from the EFW area 'freedom to trade internationally' because the latter focusses on foreign trade and does not include FDI. Both variables are only weakly correlated ($r = 0.17$).

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$$P_{i,s,c} = \alpha + \beta E_c + \sum_{j=1}^{p} \gamma_j V_{j,c} + \sum_{k=1}^{q} \delta_k W_{k,s,c} + \sum_{l=1}^{r} \zeta_l X_{l,i,s,c} + \varepsilon_{i,s,c} \quad (1)$$

First stage:

$$E_{c,t} = \eta + \sum_{m=1}^{5} \theta(E_{c,t-m} - E_{c,t-m-1}) + \sum_{j=1}^{p} \lambda_j V_{j,c} + \sum_{k=1}^{q} \mu_k W_{k,s,c} + \sum_{l=1}^{r} \upsilon_l X_{l,i,s,c} + \omega_{i,s,c} \quad (2)$$

$P_{i,s,c}$ is our educational quality variable measuring the PISA score of individual i in school s and country c. E_c represents 'economic freedom' and V a vector of p country-level control variables. W denotes a vector of q school-level control variables and X is a vector of r individual-level control variables. While α and η are the constants, $\varepsilon_{i,s,c}$ and $\omega_{i,s,c}$ are the error terms. As 'economic freedom' is a country-level variable, we cluster standard errors at this level.[2]

Equation 2 shows that our instruments are lagged differences of the instrumented variable covering the years t–1 to t–6. We perform various tests to check whether they are valid. To test for underidentification, we apply the Kleibergen and Paap (2006) *rk* LM test. Its null hypothesis is that the structural equation is underidentified. Tables 1–3 and 9 report the p-value from this test. The values suggest that the null is rejected throughout – i.e., each structural equation is identified.

As a weak instrument test, we report the Kleibergen and Paap (2006) *rk* Wald F statistic. As Baum et al. (2007) point out, this test statistic is appropriate when standard errors and statistics are cluster-robust, which is the case here. As reported in Tables 1–3 and 9, the Kleibergen-Paap *rk* Wald F statistic is well above Staiger and Stock's (1997) threshold of 10 in each of our 2SLS models. It is also well above the relevant critical value of 10.83 identified by Stock and Yogo (2005) at 10% maximal IV relative bias. This suggests that our instruments are relevant.

Finally, we test whether our instruments are exogenous. To do so, we apply Hansen's (1982) test of overidentifying restrictions. The null hypothesis of this test is that all excluded instruments are exogenous, meaning that they are uncorrelated with the error term of the second-stage regression. As can be seen in Tables 1–3 and 9, we fail to reject this hypothesis in each of our 2SLS models. Taken together, the results of the various tests suggest that our instruments are valid.

5.2 | Results

Before we discuss the results of the multivariate regressions, let us briefly take a look at the bivariate association between economic freedom and PISA scores at the country level (Figure 1). The association is moderately positive. The R^2 of the underlying regression suggests that economic freedom alone accounts for 31% of the cross-country variation in achievement.

Table 1 reports our main estimates. While column 1 presents the results for the full sample, columns 2 and 3 present the results for females and males, respectively. Controlling for numerous other (potential) determinants, they suggest that economic freedom has a positive effect on students' PISA scores, corroborating our hypothesis. Apparently, females benefit more than males; the absolute size of the coefficient of 'economic freedom' is larger in their regression than in the regression for males. Moreover, whereas their coefficient is statistically significant at the 1% level, the coefficient for males is significant at the 10% level only. The estimates suggest that the magnitudes of the effects are substantial: a one standard deviation increase in 'economic freedom' is associated with learning gains of 56% of a school year in the full sample, 63% among females, and 50% among males.

[2]Although the PISA dataset has a hierarchical structure, we follow common practice in economics not to use multilevel modelling but rather instrumental variables and cluster-robust standard errors to establish causality. Multilevel models were not designed for causal inference and usually do not account for endogeneity (Gelman, 2006; Hill, 2013; Oshchepkov & Shirokanova, 2022).

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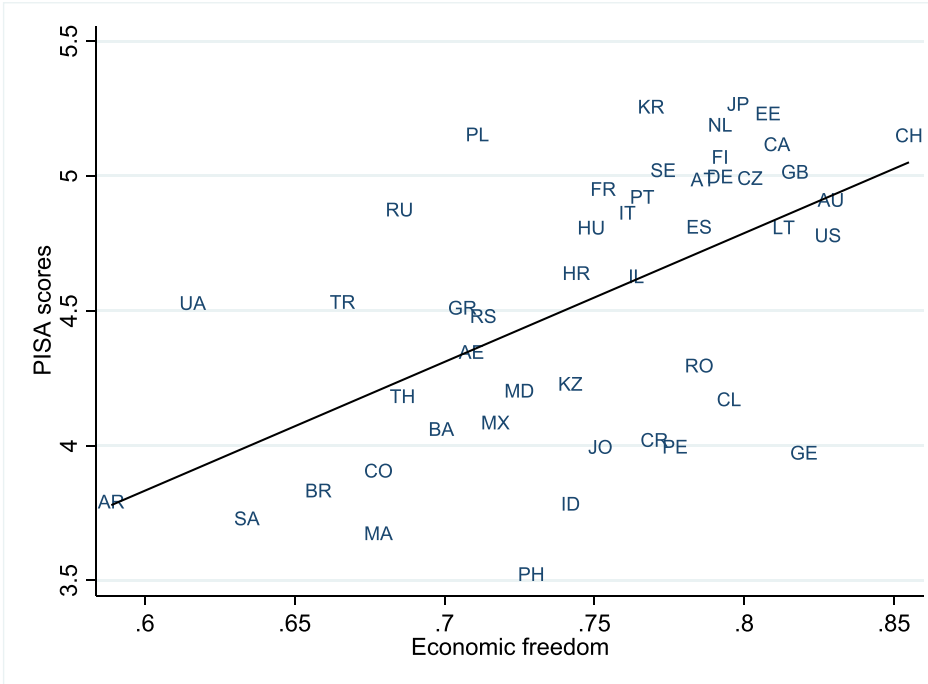

FIGURE 1 Economic freedom and PISA scores. *Note:* 47 countries. Data from 2018. Economic freedom values scaled to range from 0 to 1. PISA scores are for mathematics; original values are divided by 100. Higher values indicate more economic freedom and higher test scores, respectively. The regression represented by the fitted line yields a coefficient of 'economic freedom' of 4.78 (robust standard error = 0.94), $N = 47$, $R^2 = 0.31$. *Sources:* Gwartney et al. (2022), OECD (2019). [Colour figure can be viewed at wileyonlinelibrary.com]

Table 2 reports the results of the robustness checks. (The sample covers both females and males.) For brevity, the estimates of the standard controls are excluded; each of the models in Table 2 additionally uses the same control variables as the baseline model of Table 1. Across all robustness checks, the coefficient of 'economic freedom' is statistically significant and similar in size to the corresponding baseline estimate (Table 1, column 1), suggesting that the effect of economic freedom on the quality of education is robust.

Table 3 reports the results of two alternative specifications. The first uses OLS rather than 2SLS, as before with data from the 2018 wave. Reassuringly, the estimate of 'economic freedom' does not differ considerably from the corresponding estimate in the baseline model (Table 1, column 1). This supports our previous assessment that our instruments are valid. The fact that the coefficient of 'economic freedom' is smaller in the OLS model suggests that the variable is measured with some error. Such measurement error biases the OLS estimate downward without affecting the 2SLS estimate. As the difference in magnitude is small, it seems to be a minor issue though.

Column 2 of Table 3 reports the results of a 2SLS model that uses data from all PISA waves from 2000 to 2018.[3] Here, the coefficient of 'economic freedom' is only significant at the 10% level and much smaller in size, compared with the baseline model (Table 1, column 1). This is because data for earlier years covers almost exclusively advanced economies, for two reasons. First, far fewer developing countries took part in the early PISA waves, compared with the 2018 wave. Second, data on some control variables is unavailable for most developing countries before 2018. As a result, there is less variation in economic freedom between countries before 2018. Variation

[3]As the extended PISA dataset is not a panel dataset, a panel data estimator could not be applied. We also had to refrain from including country dummies; they would mask the effect of economic freedom because most of the variation is between countries, rather than within them.

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TABLE 1 Main estimates for PISA scores.

	(1) Full sample	(2) Females	(3) Males
Economic freedom	1.879**	2.092***	1.670*
	(0.797)	(0.721)	(0.896)
Confucian heritage society	0.507***	0.466***	0.549***
	(0.097)	(0.081)	(0.114)
GDP per capita	0.225***	0.213***	0.237**
	(0.079)	(0.072)	(0.089)
Government funding for schools	0.369	0.339*	0.394
	(0.251)	(0.195)	(0.311)
Private schools	−0.389**	−0.334**	−0.444*
	(0.183)	(0.152)	(0.220)
Central exit exams	0.001	0.044	−0.042
	(0.057)	(0.056)	−(0.061)
School autonomy	0.084	0.065	0.104
	(0.118)	(0.118)	(0.123)
Certified teachers	0.087*	0.080*	0.096*
	(0.049)	(0.047)	(0.052)
Shortage of educational material	−0.103***	−0.090***	−0.117***
	(0.021)	(0.020)	(0.023)
School location: Town	0.155***	0.137***	0.171***
	(0.028)	(0.029)	(0.028)
School location: City	0.228***	0.202***	0.252***
	(0.032)	(0.034)	(0.032)
School size	0.045*	0.034	0.055*
	(0.026)	(0.023)	(0.031)
Individual-level controls	Yes	Yes	Yes
Number of observations	408,589	204,000	204,589
Number of countries	47	47	47
Centered R^2	0.31	0.32	0.30
Kleibergen-Paap *rk* LM statistic (*p* value)	0.01	0.01	0.01
Kleibergen-Paap *rk* Wald *F* statistic	26.18	27.34	25.06
Hansen *J* statistic (*p* value)	0.60	0.59	0.61

Note: Second-stage regression results from two-stage least squares estimation, weighted by students' sampling probability. Dependent variable: 'PISA scores' (mathematics). The full sample covers both females and males. 'Economic freedom' is instrumented; the excluded instruments are lagged differences of this variable covering the years t–1 to t–6. Data from 2018, except for the excluded instruments, whose sample period stretches back to 2012. Individual-level controls: age, migration status, parental education, parental occupation, books at home, computer for school work at home, and other languages than test language spoken at home. Regression 1 additionally includes a dummy for females. All regressions also include a constant term. The critical value from the Stock-Yogo weak identification test at 10% maximal IV relative bias is 10.83. Robust standard errors, adjusted for clustering at the country level, are reported in parentheses.
***$p < 0.01$, **$p < 0.05$, and *$p < 0.10$.

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TABLE 2 Robustness checks for PISA scores.

	(1) Political freedom added	(2) Patience and risk-taking added	(3) Air pollution added	(4) Output gap added	(5) Youth unemployment rate added	(6) FDI stock added
Economic freedom	1.663**	1.761**	1.825**	1.463**	1.869**	1.745**
	(0.799)	(0.658)	(0.754)	(0.689)	(0.816)	(0.791)
Political freedom	0.181					
	(0.126)					
Patience		0.298**				
		(0.114)				
Risk-taking		−0.603***				
		(0.102)				
Air pollution			−0.815**			
			(0.327)			
Output gap				8.835**		
				(3.915)		
Youth unemployment rate					−0.027	
					(0.292)	
FDI stock						0.049*
						(0.027)
Main control variables	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	408,589	408,589	408,589	408,589	408,589	408,589
Number of countries	47	47	47	47	47	47
Centered R^2	0.31	0.32	0.31	0.31	0.31	0.31
Kleibergen-Paap *rk* LM statistic (*p* value)	0.03	0.01	0.05	0.01	0.01	0.01
Kleibergen-Paap *rk* Wald *F* statistic	21.19	31.04	28.55	22.89	25.38	26.06
Hansen *J* statistic (*p* value)	0.62	0.69	0.24	0.78	0.60	0.57

Note: Second-stage regression results from two-stage least squares estimation, weighted by students' sampling probability. Dependent variable: 'PISA scores' (mathematics). 'Economic freedom' is instrumented with lagged differences of this variable, covering the years $t-1$ to $t-6$. Data from 2018, except for the excluded instruments, whose sample period stretches back to 2012. In addition to the explanatory variables mentioned above, each regression additionally uses the same control variables as the baseline regressions (Table 1). All regressions also include a constant term. The sample covers both females and males. The critical value from the Stock-Yogo weak identification test at 10% maximal IV relative bias is 10.83. Robust standard errors, adjusted for clustering at the country level, are reported in parentheses.
***$p < 0.01$, **$p < 0.05$, and *$p < 0.10$.

within countries from 2000 to 2018 is even smaller. Both of these facts support our decision to use only 2018 data in our main model.

Finally, some brief remarks on the new controls (Tables 1–3). In line with expectations, 'Confucian heritage society' has a statistically highly significant and positive effect on PISA results. Moreover, we find that 'economic freedom' does not proxy for 'political freedom' and that the latter does not have any effect. While our results for

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TABLE 3 Supplementary estimates for PISA scores.

	(1) OLS rather than 2SLS	(2) All waves rather than 2018 wave only
Economic freedom	1.639**	1.085*
	(0.695)	(0.633)
Confucian heritage society	0.515***	0.637***
	(0.096)	(0.122)
GDP per capita	0.232***	0.281***
	(0.082)	(0.088)
Government funding for schools	0.397	0.331*
	(0.253)	(0.190)
Private schools	−0.384**	−0.361*
	(0.185)	(0.199)
Central exit exams	0.001	0.033
	(0.058)	(0.065)
School autonomy	0.092	0.045
	(0.116)	(0.108)
Certified teachers	0.088*	0.082
	(0.048)	(0.056)
Shortage of educational material	−0.105***	−0.091***
	(0.021)	(0.018)
School location: Town	0.153***	0.067**
	(0.028)	(0.028)
School location: City	0.225***	0.122***
	(0.034)	(0.034)
School size	0.045*	0.067***
	(0.026)	(0.022)
Individual-level controls	Yes	Yes
Number of observations	408,589	1,987,478
Number of countries	47	49
Centered R^2	0.31	0.33
Kleibergen-Paap *rk* LM statistic (*p* value)	n/a	0.01
Kleibergen-Paap *rk* Wald *F* statistic	n/a	1,533.92
Hansen *J* statistic (*p* value)	n/a	0.47

Note: Dependent variable: 'PISA scores' (mathematics). While column 1 reports the results from an OLS regression that uses 2018 data, column 2 reports the results from a two-stage least squares (2SLS) model based on all PISA waves up to 2018 (2000, 2003, 2006, 2009, 2012, 2015, 2018). In the 2SLS model, 'economic freedom' is instrumented with lagged levels of 'economic freedom', covering the years $t-1$ and $t-2$. In all other respects, both models correspond to the baseline model of Table 1, column 1. The sample covers both females and males. Individual-level controls: female, age, migration status, parental education, parental occupation, books at home, computer for school work at home, and other languages than test language spoken at home. The 2SLS model additionally includes wave dummies. All regressions also include a constant term. In the 2SLS model, the critical value from the Stock-Yogo weak identification test at 10% maximal IV size is 19.93. Robust standard errors, adjusted for clustering at the country level, are reported in parentheses.
***$p < 0.01$, **$p < 0.05$, and *$p < 0.10$.

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'patience' and 'risk-taking' chime with Hanushek et al. (2022), our result for 'air pollution' suggests that its adverse impact on learning outcomes is not just a local problem in a few countries but a nationwide problem in many.

6 | ANALYZING WORLD BANK SCORES

6.1 | Data and methodology

To construct harmonized test scores for a large number of countries, the World Bank has used three major international testing programs: PISA, the Trends in International Mathematics and Science Study (TIMSS) program, and the Progress in International Reading Literacy Study (PIRLS). It has combined them with four major regional testing programs: the Latin American Laboratory for Assessment of the Quality of Education (LLECE), the Southern and Eastern Africa Consortium for Monitoring Educational Quality (SACMEQ), the Pacific Islands Learning and Numeracy Assessment (PILNA) and the Program for the Analysis of Education Systems (PASEC). Moreover, it has incorporated the Early Grade Reading Assessments (EGRAs) coordinated by the US Agency for International Development (World Bank, 2023). Patrinos and Angrist (2018) have developed a harmonization methodology that can be used to place the test scores from the various programs on a common scale. As the harmonized scores combine tests of student competences in several fields (particularly, reading, mathematics, and science), they measure achievement comprehensively. The scores are scaled to range from 0 (minimal attainment) to 100 (advanced attainment).

As in our PISA regressions, we control for both 'Confucian heritage society' and 'GDP per capita' throughout. However, in contrast to these regressions, we now have only one control for national school systems: 'government expenditure on education'. This is for three reasons. First, we have no school-level data for our World Bank regressions. Second, the various country-level measures of school systems included in our PISA regressions are available for a subset of PISA countries only. Third, other country-level measures of school systems are available for a few countries too. All of this does not seem to be a major drawback though because 'government expenditure on education' is a reasonably good overall proxy of national school systems and most measures of national school systems included in our PISA regressions are not statistically significant at conventional levels anyway.

As explained in section 2, economic freedom is likely to raise GDP per capita and may also increase the share of government education expenditure in GDP. Thus some of the effect of economic freedom on educational quality may run through these variables. Section 7 investigates these indirect effects. For the same reasons as in section 5, we do not omit these variables here: first, they are not exclusively determined by economic freedom and, second, they are likely to have a direct effect on educational outcomes.

Instead of the school location dummies, which are highly significant in our PISA regressions, we now use the urbanization rate because countries with higher levels of urbanization have more schools in towns and cities. A further reason to control urbanization is that access to school is usually better in urban than in rural areas (e.g., Bertinelli & Zou, 2008). Thus a higher urbanization rate could bolster the quality of education.

Our next control, also used throughout, is the share of children in the population. Both parents and societies face a trade-off between child quantity and quality: the higher the number of children, the less they are usually able to invest in each of them (e.g., Becker & Lewis, 1973; Hanushek, 1992). Thus a higher share of children in the population could result in a lower level of educational quality. Finally, in our robustness checks we add the same variables as in our PISA robustness checks, and for the same reasons.

Data for World Bank test scores is available for 2017, 2018, 2020 and, for some countries, for 2010 too. However, in our baseline specification, we use 2018 data only, for three reasons. First, it improves the comparability of our estimates with those from our main PISA regressions. Second, as we have data from only a few years a fixed effects estimator is not an option. Third, random effects estimates are not an option either because they fail the Hausman (1978) test. Still, in a supplementary model, we do use data from all years (see below).

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As with the PISA data, we mainly run two-stage least squares regressions, instrumenting 'economic freedom'. The baseline specification is as follows:

Second stage:

$$T_c = \pi + \kappa E_c + \sum\nolimits_{n=1}^{s} \chi_n Z_{n,c} + o_c \tag{3}$$

First stage:

$$E_{c,t} = \rho + \tau E_{c,t-1} + \upsilon E_{c,t-2} + \sum\nolimits_{n=1}^{s} \varphi_n Z_{n,c,t} + \psi_{c,t} \tag{4}$$

T_c is the World Bank test score of country c, our measure of educational quality. E denotes 'economic freedom' and Z is a vector of s control variables. While π and ρ are the constants, o_c and $\psi_{c,t}$ are the error terms.

As explained in section 4, in our regressions with World Bank test scores the excluded instruments of 'economic freedom' are lagged levels of this variable from the previous two years (equation 4). Tables 4–8 report the results of the IV validity tests. While the Kleibergen-Paap *rk* LM statistic indicates that each structural equation is identified, the first-stage *F* statistic and Hansen's *J* test suggest that our instruments are relevant and exogenous, and thus valid.

TABLE 4 Main estimates for World Bank test scores.

	(1) Full sample	(2) Females	(3) Males
Economic freedom	53.96***	46.20***	53.09***
	(16.14)	(16.28)	(18.99)
Confucian heritage society	14.27***	13.31***	14.38***
	(4.03)	(4.08)	(4.05)
GDP per capita	6.57**	7.02**	6.00*
	(2.67)	(2.70)	(3.09)
Government expenditure on education	1.12*	1.21*	1.08
	(0.64)	(0.72)	(0.79)
Urbanization rate	−12.69*	−13.63	−12.30
	(7.56)	(8.55)	(9.16)
Child population share	−69.52***	−79.89***	−77.01***
	(23.11)	(23.32)	(26.69)
Number of observations & countries	132	120	120
Centered R^2	0.72	0.76	0.70
Kleibergen-Paap *rk* LM statistic (*p* value)	0.00	0.00	0.00
F test of excluded instruments (first stage)	745.98	623.20	623.20
Hansen *J* statistic (*p* value)	0.83	0.89	0.83

Note: Second-stage regression results from two-stage least squares estimation. Dependent variables: 'World Bank test scores, full sample' (column 1); 'World Bank test scores, females' (column 2) and 'World Bank test scores, males' (column 3). The full sample covers both females and males. 'Economic freedom' is instrumented; the excluded instruments are lagged levels of 'economic freedom', covering the years t–1 and t–2. Data from 2018, except for the excluded instruments, whose data are from 2016 and 2017. All regressions also include a constant term. The critical value from the Stock-Yogo weak identification test at 10% maximal IV size is 19.93. Robust standard errors are reported in parentheses.
***p < 0.01, **p < 0.05, and *p < 0.10.

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TABLE 5 Robustness checks for World Bank test scores.

	(1) Political freedom added	(2) Patience and risk-taking added	(3) Air pollution added	(4) Output gap added	(5) Youth unemployment rate added	(6) FDI stock added
Economic freedom	57.45***	59.81***	38.10**	61.98***	43.63***	54.30***
	(17.52)	(20.76)	(15.93)	(15.89)	(16.00)	(16.22)
Political freedom	−1.86					
	(5.45)					
Patience		7.24				
		(5.64)				
Risk-taking		−10.94*				
		(6.38)				
Air pollution			−21.39***			
			(6.05)			
Output gap				−1.27		
				(1.47)		
Youth unemployment rate					−24.57**	
					(11.71)	
FDI stock						−5.49
						(5.82)
Main control variables	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations & countries	132	68	131	132	132	132
Centered R^2	0.72	0.74	0.74	0.73	0.74	0.72
Kleibergen-Paap *rk* LM statistic (*p* value)	0.00	0.00	0.00	0.00	0.00	0.00
F test of excluded instruments (first stage)	706.54	472.98	728.09	803.63	853.80	745.58
Hansen *J* statistic (*p* value)	0.85	0.81	0.84	0.80	0.79	0.84

Note: Second-stage regression results from two-stage least squares estimation. Dependent variable: 'World Bank test scores, full sample' (females and males). 'Economic freedom' is instrumented with lagged levels of 'economic freedom', covering the years $t-1$ and $t-2$. Data from 2018, except for the excluded instruments, whose data are from 2016 and 2017. In addition to the explanatory variables mentioned above, each regression additionally uses the same control variables as the baseline regressions (Table 4). All regressions also include a constant term. The critical value from the Stock-Yogo weak identification test at 10% maximal IV size is 19.93. Robust standard errors are reported in parentheses.
***p < 0.01, **p < 0.05, *p < 0.10.

6.2 | Results

Figure 2 plots the World Bank test scores against the economic freedom ratings for our sample of 132 countries. Generally, higher levels of economic freedom are correlated with higher test scores. According to the R^2 of the underlying regression, economic freedom accounts for 49% of the variance in test scores across countries, an even stronger association than for the 47 PISA countries in Figure 1.

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Table 4 shows our main estimates. Although we use just five controls, the regressions explain 72% of the variance in test scores for the full sample of females and males and even 76% for females. The coefficient of 'economic freedom' is positive and highly statistically significant in each of the three regressions, suggesting that economic freedom improves educational outcomes. This corroborates our hypothesis and is in line with our PISA estimates. In our large sample of countries, males benefit slightly more than females.

Our estimates suggest that the size of the effect is substantial. For example, take New Zealand and Brazil. Whereas New Zealand had one of the highest degrees of economic freedom and one of the best World Bank test scores, Brazil's values were well below the sample mean in both areas (Figure 2). According to our estimates, if economic freedom in Brazil had been as high as in New Zealand, its World Bank score might have been 31% higher, ceteris paribus. Of course, any such comparison between countries should be treated with caution. Still, the figure suggests that economic freedom could have a substantial effect on the quality of education.

In our robustness checks (Table 5), the coefficient of 'economic freedom' is similar to the one from the corresponding baseline regression (Table 4, column 1). The coefficient of 'political freedom' remains statistically insignificant, as in the corresponding PISA regression. Interestingly, 'patience' becomes insignificant and 'risk-taking' significant at the 10% level only. Whereas 'output gap' and 'FDI stock' become insignificant too, the coefficient of 'air pollution' remains statistically highly significant and negative. The estimate of 'youth unemployment rate' is now

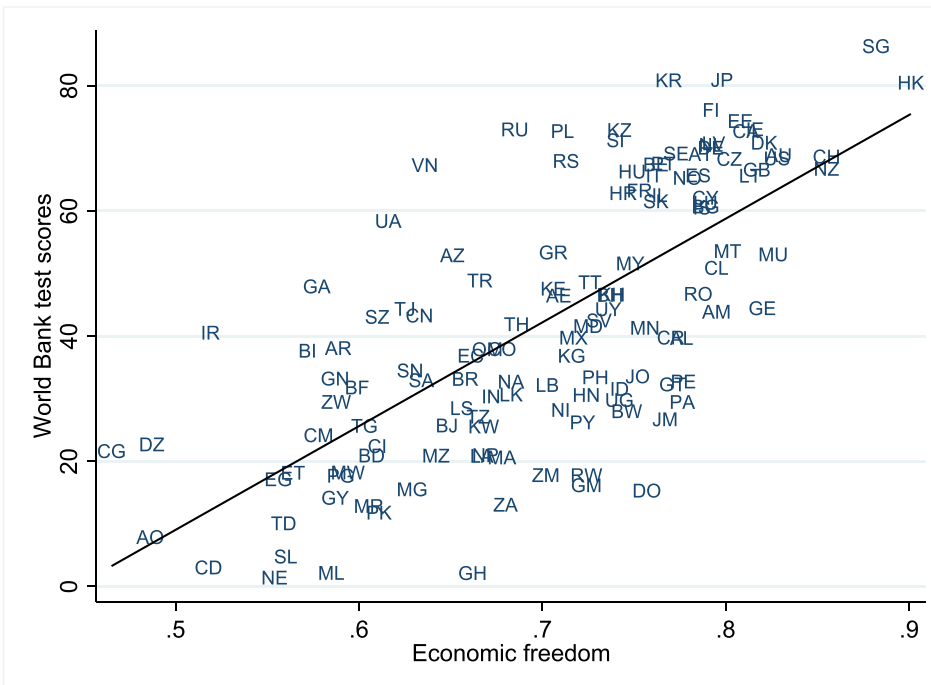

FIGURE 2 Economic freedom and World Bank test scores. *Note:* 132 countries. Data from 2018. 'Economic Freedom of the World' summary ratings, scaled to range from 0 (least free) to 1 (most free). Harmonized test scores constructed by the World Bank, based on several multi-country student achievement testing programs such as TIMSS (Trends in International Mathematics and Science Study), PIRLS (Progress in International Reading Literacy Study), and PISA (Programme for International Student Assessment). The test scores are scaled to range from 0 (minimal attainment) to 100 (advanced attainment). They cover both females and males. The regression represented by the fitted line yields a coefficient of 'economic freedom' of 165.73 (robust standard error = 13.37), $N = 132$, $R^2 = 0.49$. *Sources:* Gwartney et al. (2022), World Bank (2022). [Colour figure can be viewed at wileyonlinelibrary.com]

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also significant, suggesting that higher youth unemployment might reduce learning outcomes. Most importantly, as in our PISA regressions, the results reported in Table 5 indicate that the positive effect of economic freedom on educational quality is robust.

Table 6 reports the results of our two alternative specifications. The first substitutes OLS for 2SLS, with the sample period remaining in 2018. As in the analogous PISA exercise, the OLS estimate of 'economic freedom' is slightly below the corresponding baseline estimate (Table 4, column 1), suggesting that there is some minor measurement error that biases the OLS estimate downward. In general, though, the results of the OLS estimates accord with our 2SLS estimates.

Column 2 of Table 6 presents the results of a model that includes data from all available years – i.e., 2010, 2017, 2018, and 2020. To control for both endogeneity and unobserved country effects, we employ a 2SLS random effects estimator. The coefficient of 'economic freedom' is significant at the 1% level and only slightly smaller than in the corresponding baseline model (Table 4, column 1). This suggests that economic freedom might have had a positive effect on educational achievement over the extended sample period. However, the model fails to meet the asymptotic assumptions of the Hausman (1978) test.

TABLE 6 Supplementary estimates for World Bank test scores.

	(1) OLS rather than 2SLS	(2) All years rather than 2018 only
Economic freedom	53.16***	43.72***
	(15.49)	(14.87)
Confucian heritage society	14.27***	14.17***
	(4.03)	(4.18)
GDP per capita	6.60**	6.71***
	(2.65)	(2.29)
Government expenditure on education	1.12*	−0.00
	(0.64)	(0.35)
Urbanization rate	−12.71*	−8.49
	(7.58)	(6.86)
Child population share	−69.77***	−63.85***
	(23.25)	(18.83)
Number of observations	132	483
Number of countries	132	137
Centered R^2	0.72	0.70
Kleibergen-Paap *rk* LM statistic (*p* value)	n/a	0.00
Kleibergen-Paap *rk* Wald *F* statistic	n/a	672.93
Hansen *J* statistic (*p* value)	n/a	0.70

Note: Dependent variable: 'World Bank test scores, full sample' (females and males). While column 1 reports the results from an OLS regression that uses 2018 data, column 2 reports the results from a two-stage least squares (2SLS) model that uses data from all available years (2010, 2017, 2018, 2020). In the 2SLS model, 'economic freedom' is instrumented with lagged levels of 'economic freedom', covering the years $t-1$ and $t-2$. This model uses both country random effects and year dummies. In all other respects, both models correspond to the baseline model of Table 4, column 1. All regressions also include a constant term. In the 2SLS random effects model, the critical value from the Stock-Yogo weak identification test at 10% maximal IV size is 19.93. The model fails to meet the asymptotic assumptions of the Hausman (1978) test. Robust standard errors are reported in parentheses.
***$p < 0.01$, **$p < 0.05$, and *$p < 0.10$.

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7 | TYPES OF IMPACT

This section investigates the various ways in which economic freedom is likely to affect the quality of education. First, it probably increases parents' prioritization of education – mainly because it enhances the returns to educational investments (King et al., 2012) but also because it provides them with more choices among schools and makes it easier for them to fund their children's education (section 2). Economic freedom probably strengthens educational prioritization among policymakers too – not only to meet parents' demand for quality education but also because higher educational returns entail benefits for the economy and society at large.

Using World Values Survey data on 48 countries, a previous paper of ours finds that, indeed, people in economically freer countries attach greater importance to education (Feldmann, 2021). Figure 3 summarizes its main results. Controlling for respondents' demographic and social characteristics and their other values as well as for unobserved country effects, it finds that a 10 percentage point increase in economic freedom raises the probability that they were worried about education in their own country by an average of 5.3 percentage points. Furthermore, it finds that economic freedom raises the probability that parents were worried about the education of their children by an average of 4.2 percentage points, ceteris paribus. These stronger educational concerns very likely prompt both parents and policymakers to take actions that improve educational outcomes.

Next, let us look at the individual areas of economic freedom. This provides further insights into the ways in which economic freedom affects the quality of education. Table 7 reports the results of 2SLS regressions in which

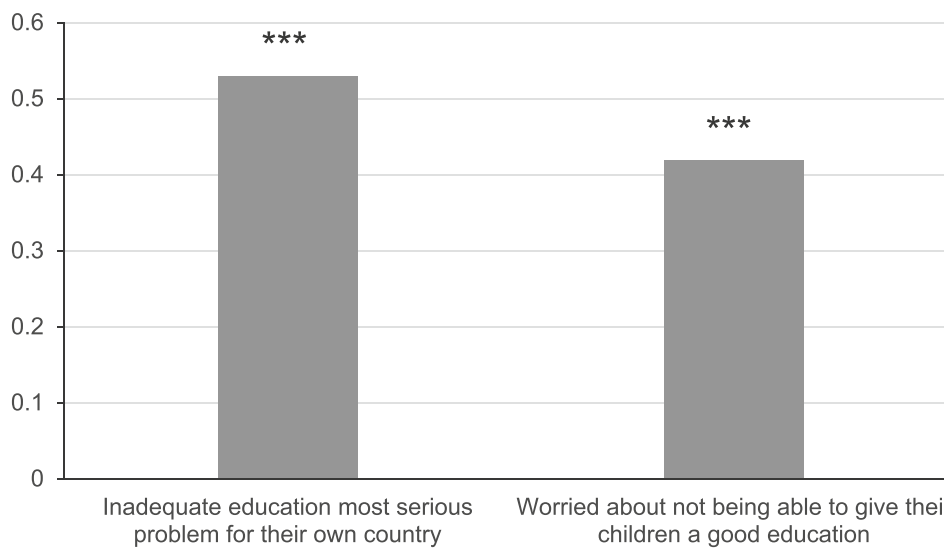

FIGURE 3 Effects of economic freedom on survey respondents' educational concerns. *Note:* average marginal effects of probit regressions. The first regression uses as a dependent variable a dummy that equals one if a respondent selected as their first choice "inadequate education" as the most serious problem for their own country. The alternatives were: "people living in poverty and need", "discrimination against girls and women", "poor sanitation and infectious diseases" and "environmental pollution". This question was included in wave 5 (2005–09) of the World Values Survey. The second regression uses as a dependent variable a dummy that equals one if a respondent said that they worried very much or a great deal about "not being able to give my children a good education". It equals zero if they said that they worried about this not much or not at all. This question was included in wave 6 (2010–14) of the World Values Survey. 'Economic freedom' is the EFW summary index, scaled to range from 0 (least free) to 1 (most free). Both regressions control for respondents' demographic and social characteristics, for their other values, and for unobserved country effects. While regression 1 uses 14,999 observations from 24 countries, regression 2 uses 26,222 observations from 33 countries. ***$p < 0.01$. *Source:* Feldmann (2021).

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TABLE 7 Areas of economic freedom, World Bank test scores.

	(1)	(2)	(3)	(4)	(5)
Size of government	−11.13				
	(11.60)				
Legal system & property rights		48.19***			
		(10.53)			
Sound money			25.54**		
			(11.33)		
Freedom to trade internationally				30.19***	
				(10.28)	
Regulation					53.58***
					(12.42)
Main control variables	Yes	Yes	Yes	Yes	Yes
Number of observations & countries	132	132	132	132	132
Centered R^2	0.70	0.74	0.71	0.71	0.74
Kleibergen-Paap *rk* LM statistic (*p* value)	0.00	0.00	0.00	0.00	0.00
F test of excluded instruments (first stage)	710.49	1,953.15	202.39	155.32	643.74
Hansen *J* statistic (*p* value)	0.70	0.58	0.37	0.97	0.33

Note: Second-stage regression results from two-stage least squares estimation. Dependent variable: 'World Bank test scores, full sample' (females and males). Each of the five economic freedom area variables is instrumented with its lagged levels, covering the years $t-1$ and $t-2$. Data from 2018, except for the excluded instruments, whose data are from 2016 and 2017. Each regression additionally uses the same control variables as the baseline regressions (Table 4). All regressions also include a constant term. The critical value from the Stock-Yogo weak identification test at 10% maximal IV size is 19.93. Robust standard errors are reported in parentheses.
***$p < 0.01$, **$p < 0.05$, and *$p < 0.10$.

we replace the EFW summary index with the area indices, one at a time. The regressions use our World Bank sample of 132 countries and the model we applied in section 6 to analyze it. Thus each area variable is instrumented with its lagged values from the previous two years. The estimates show that four of the five areas have a significant effect on student achievement. Although we must be careful in interpreting these results, we can tentatively draw the following conclusions. The estimate of 'regulation', which has the largest coefficient, suggests that market-friendly regulation is conducive to educational quality. This might be because flexible credit market regulations make it easier for parents to fund quality education for their children, or because flexible business regulations provide a large scope for private schools, or because flexible labor market and business regulations increase the availability of high-skill jobs. Free trade can incentivize investment in quality education because many export-oriented firms require highly skilled workers (Baskaran & Hessami, 2012). Secure property rights, a strong rule of law, and monetary stability affect educational quality positively as well, according to our estimates (Table 7). As explained in section 2, this is probably because these freedoms secure people's real returns to educational investments. Interestingly, the size of the government has no effect.[4] Perhaps, a negative effect of high taxes is offset by a positive effect of high spending on education.

As mentioned in section 2, economic freedom is also likely to improve educational quality by raising economic growth and income per capita. Thus the positive effects of 'GDP per capita' on both World Bank scores (Table 4) and PISA scores (Table 1) are probably partly due to economic freedom's positive effect on GDP per capita. While

[4]The fact that government size is the only area not to affect educational performance supports Ott (2018) and Bergh (2020), who argue that this area is much less important for economic freedom than the other four.

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TABLE 8 Indirect effects added, World Bank test scores.

	(1) Indirect effect via 'GDP per capita' added	(2) Indirect effect via 'government expenditure on education' added
Economic freedom	62.99***	56.31***
	(15.61)	(16.09)
Confucian heritage society	13.11***	13.07***
	(4.03)	(3.93)
GDP per capita	6.60**	6.61**
	(2.64)	(2.66)
Government expenditure on education	1.14*	1.12*
	(0.64)	(0.64)
Urbanization rate	−0.85	−11.17
	(6.44)	(7.61)
Child population share	−112.73***	−69.40***
	(14.48)	(23.10)
Number of observations & countries	132	132
Centered R^2	0.72	0.72
Kleibergen-Paap *rk* LM statistic (*p* value)	0.00	0.00
F test of excluded instruments (first stage)	818.10	716.98
Hansen *J* statistic (*p* value)	0.83	0.83

Note: Second-stage regression results from two-stage least squares estimation. Dependent variable: 'World Bank test scores, full sample' (females and males). 'Economic freedom' is instrumented with lagged levels of 'economic freedom', covering the years $t-1$ and $t-2$. Data from 2018, except for the excluded instruments, whose data are from 2016 and 2017. In model 1, 'GDP per capita' is replaced by the residuals from a regression in which this variable is used as dependent variable. In model 2, 'government expenditure on education' is replaced by the residuals from a regression in which this variable is used as dependent variable. All other control variables are the same as in the baseline model (Table 4). All regressions also include a constant term. The critical value from the Stock-Yogo weak identification test at 10% maximal IV size is 19.93. Robust standard errors are reported in parentheses.
***$p < 0.01$, **$p < 0.05$, and *$p < 0.10$.

we disregard these indirect effects in sections 5 and 6, we study them here using a method suggested by Gwartney et al. (2006). We first estimate equations with 'GDP per capita' as the dependent variable and 'economic freedom' and the baseline controls as right-hand side variables. Subsequently, we substitute the residuals from these regressions for 'GDP per capita' in our baseline specifications. The logic of doing this is that the residuals from the 'GDP per capita' regressions represent the variation that is uncorrelated with 'economic freedom'. By using these residuals, the variation in 'GDP per capita' that is associated with differences in 'economic freedom' is captured in the coefficients of 'economic freedom'. Thus these coefficients reflect both the direct effect as well as the indirect effect via 'GDP per capita'. The results are reported in column 1 of Table 8 for World Bank scores and column 1 of Table 9 for PISA scores. As expected, the coefficients of 'economic freedom' are larger than the coefficients in the corresponding baseline models (Table 4, column 1 and Table 1, column 1). With the indirect path through 'GDP per capita' additionally taken into account, Brazil's World Bank test score might have been 36% higher, had its economic freedom rating been as high as New Zealand's, ceteris paribus. The result of the modified PISA regression indicates that a one standard deviation increase in 'economic freedom' is associated with learning gains of 68% of a school year. Both of these estimates are noticeably larger than the estimates of the corresponding baseline models (sections 5 and 6).

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TABLE 9 Indirect effects added, PISA scores.

	(1) Indirect effect via 'GDP per capita' added	(2) Indirect effect via 'government funding for schools' added	(3) Other transmission variables omitted
Economic freedom	2.271***	2.210***	2.760**
	(0.753)	(0.774)	(1.226)
Confucian heritage society	0.539***	0.453***	0.482***
	(0.094)	(0.082)	(0.125)
GDP per capita	0.230***	0.220***	0.317***
	(0.080)	(0.081)	(0.096)
Government funding for schools	0.360	0.402	0.667*
	(0.252)	(0.253)	(0.396)
Private schools	−0.333*	−0.482***	
	(0.188)	(0.174)	
Central exit exams	−0.003	−0.009	−0.005
	(0.057)	(0.059)	(0.074)
School autonomy	0.228**	0.109	−0.010
	(0.109)	(0.117)	(0.140)
Certified teachers	0.115**	0.122**	
	(0.053)	(0.054)	
Shortage of educational material	−0.130***	−0.100***	
	(0.019)	(0.021)	
School location: Town	0.171***	0.157***	0.198***
	(0.029)	(0.028)	(0.040)
School location: City	0.236***	0.231***	0.304***
	(0.032)	(0.032)	(0.042)
School size	0.031	0.037	0.031
	(0.027)	(0.026)	(0.032)
Individual-level controls	Yes	Yes	Yes
Number of observations	408,589	408,589	408,589
Number of countries	47	47	47
Centered R^2	0.31	0.31	0.23
Kleibergen-Paap *rk* LM statistic (*p* value)	0.02	0.02	0.02
Kleibergen-Paap *rk* Wald *F* statistic	34.40	26.14	25.69
Hansen *J* statistic (*p* value)	0.60	0.61	0.19

Note: Second-stage regression results from two-stage least squares estimation, weighted by students' sampling probability. Dependent variable: 'PISA scores' (mathematics). 'Economic freedom' is instrumented with lagged differences of this variable, covering the years t–1 to t–6. Data from 2018, except for the excluded instruments, whose sample period stretches back to 2012. In model 1, 'GDP per capita' is replaced by the residuals from a regression in which this variable is used as dependent variable. In model 2, 'government funding for schools' is replaced by the residuals from a regression in which this variable is used as dependent variable. Model 3 excludes other variables through which economic freedom might affect PISA scores; specifically 'private schools', 'certified teachers', 'shortage of educational material', 'books at home', 'computer for school work at home'. All other control variables are the same as in the baseline model (Table 1). All regressions also include a constant term. The sample covers both females and males. The critical value from the Stock-Yogo weak identification test at 10% maximal IV relative bias is 10.83. Robust standard errors, adjusted for clustering at the country level, are reported in parentheses.
***$p < 0.01$, **$p < 0.05$, and *$p < 0.10$.

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Next, we use the same method to study the indirect effect of government education expenditure. As argued in section 2, economic freedom may prompt governments to raise this type of spending. Therefore, we now replace 'government expenditure on education' in the World Bank regressions and 'government funding for schools' in the PISA regressions with residuals from regressions in which they are the dependent variables. This captures both the direct effect of economic freedom and its indirect effect via the government spending variables. Column 2 of Table 8 and column 2 of Table 9 present the results for World Bank scores and PISA scores, respectively. In both cases, the coefficient of 'economic freedom' rises, compared with the corresponding baseline models (Table 4, column 1 and Table 1, column 1). However, the increase is smaller than in the case of 'GDP per capita'. It is almost negligible when using World Bank scores. Moreover, the direct effects of government education spending on test scores are small and insignificant at conventional levels throughout (Tables 1, 4, 8, and 9). All of this chimes with the previous literature, according to which cross-country differences in government education spending play a limited role in explaining differences in student achievement (Woessmann, 2016).

Finally, our PISA regressions include a range of variables measuring private schools and school and family resources – specifically, 'private schools', 'certified teachers', 'shortage of educational material', 'books at home', and 'computer for school work at home'. They, too, might represent channels through which 'economic freedom' affects test scores (section 2). As it is impractical to apply Gwartney et al.'s (2006) method to all of these possible further transmission variables, we exclude them in our final specification (Table 9, column 3). Although the method of omitting potential transmission variables is occasionally applied in the literature (e.g., Gründler & Scheuermeyer, 2018), it results in omitted variable bias if they are partly determined by other factors than the variable of interest and have a direct effect on the outcome variable. As this applies to the variables mentioned above (section 5.1), the estimate of 'economic freedom' from our final specification should be viewed as indicative only. Still, it is noteworthy that the coefficient of 'economic freedom' rises strongly when they are excluded (column 3 of Table 9 compared with column 1 of Table 1). This could mean that economic freedom has noticeable effects on educational quality that run through private schools and/or school and family resources.

8 | CONCLUSION

Although there is a large body of literature on the effects of economic freedom and a separate one on the causes of international differences in student achievement, our paper is the first to combine the two and study whether economic freedom affects the quality of education. We hypothesize that there might be a positive effect, for several reasons. In short, economic freedom incentivizes parents to invest in high-quality education for their children and helps them to do so. It also incentivizes and helps both governments and private providers to deliver high-quality education. Using two different indicators of educational quality and two different datasets covering large samples of countries, we find evidence to support our hypothesis. The magnitudes of the estimated effects are substantial. They are even larger once the indirect impact of economic freedom via GDP per capita and, to a lesser extent, once the indirect impact via government education expenditure is taken into account. Indirect effects via private providers and/or school and family resources might enhance the impact of economic freedom even further, according to our results.

As a high quality of education is crucial for both children's earnings prospects and macroeconomic development, our findings have important implications for public policy. While bolstering economic freedom should be a general principle of policy-making, several issues stand out. First, credit markets should be regulated pro-competitively, providing parents with a large supply of loans to fund a high-quality education for their children. Property rights should be precisely defined and secure, enabling parents to use their property as collateral. Labor markets should be regulated pro-competitively too – most importantly, to enable young people to enter professions in which they can achieve high returns on their human capital. Relatedly, business regulation should be market-friendly as well, encouraging the presence of private education providers and ensuring the availability of well-paid jobs. Jointly, a pro-

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competitive regulation of these areas plus the security of property rights incentivize and enable parents and students to invest in high-quality education.

Free trade policies are advisable from an educational-quality perspective for two reasons. First, a large export sector usually increases the availability of jobs for high-skilled workers, incentivizing parents and students to invest in high-quality education. Second, free trade raises GDP per capita, which, according to our research, has positive knock-on effects on learning outcomes.

Monetary policy, supported by an appropriate fiscal policy, should maintain low inflation. This secures the return to educational investments in real terms, also incentivizing parents and students to invest in high-quality education. Upholding the rule of law is important for educational performance in several ways, from securing people's returns to educational investments to enabling both the economy and the state to work smoothly and predictively. In this way, the rule of law incentivizes and facilitates investment in high-quality education.

Although we did not find evidence that a small size of government per se is important for student achievement, keeping the income tax burden at a moderate level is essential in order for individuals to achieve high net returns on their human capital, again incentivizing investment in quality education. On the other hand, the government needs to raise sufficient tax revenue to fund good schools. A final policy implication of our research is that elections should be free and fair so that politicians get elected who act on parents' preferences for good schools.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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